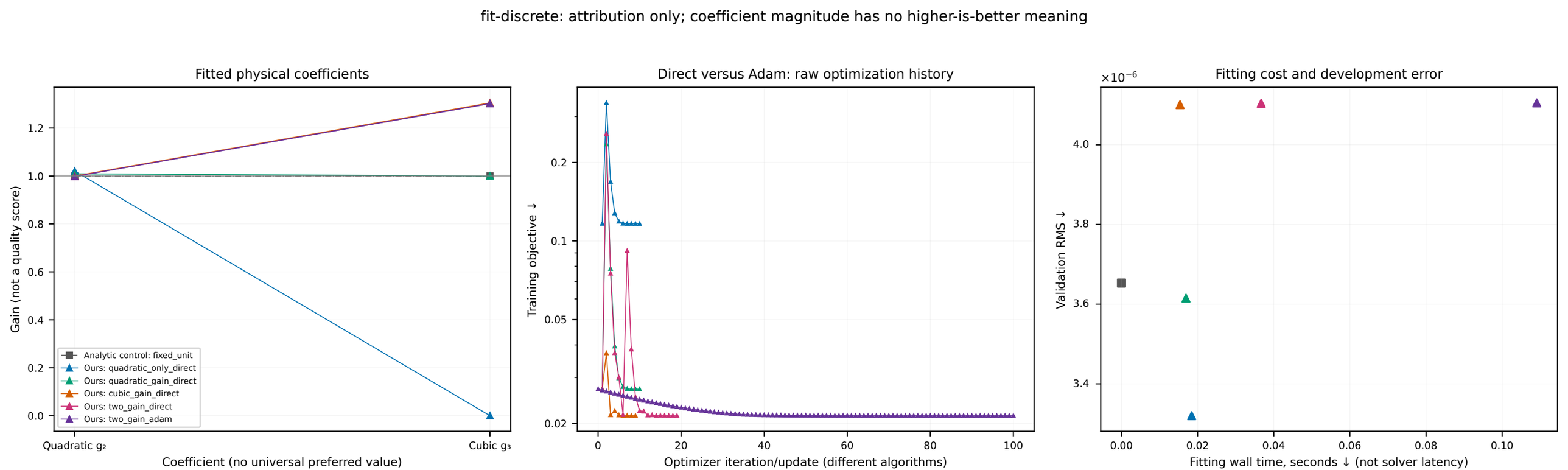
fit-discrete: attribution only; coefficient magnitude has no higher-is-better meaning
Fitted physical coefficients
Gain (not a quality score)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Analytic control: fixed_unit
Ours: quadratic_only_direct
Ours: quadratic_gain_direct
Ours: cubic_gain_direct
Ours: two_gain_direct
Ours: two_gain_adam
Quadratic g₂
Cubic g₃
Coefficient (no universal preferred value)
Direct versus Adam: raw optimization history
Training objective ↓
0.2
0.1
0.05
0.02
0
20
40
60
80
100
Optimizer iteration/update (different algorithms)
Fitting cost and development error
×10⁻⁶
Validation RMS ↓
4.0
3.8
3.6
3.4
0.00
0.02
0.04
0.06
0.08
0.10
Fitting wall time, seconds ↓ (not solver latency)
Development evidence · lower error/cost is better · observed bands ≠ confidence intervals